

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Analytical Problem-Solving Assessment

Duration: 1hr

1. A warehouse robot must determine the shortest path to transport goods while avoiding obstacles. Apply **Dynamic Programming** to develop an optimal policy and explain how value iteration can improve navigation efficiency. **(5 Marks)**
2. A recommendation system collects customer interactions over multiple sessions. Apply **Monte Carlo Control** to update the policy and explain how the agent improves recommendations with experience. **(5 Marks)**
3. A delivery robot operates in an uncertain environment. Apply the **SARSA** algorithm to update the action-value function and explain how the learned policy adapts to environmental changes. **(5 Marks)**
4. A robotic arm performs continuous object manipulation. Apply the **REINFORCE** algorithm to develop a policy that maximizes task completion accuracy. .

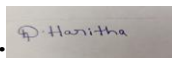
(5 Marks)

5. An autonomous vehicle is trained using **Q-Learning** and **Deep Q-Networks (DQN)**. Analyze the differences in learning performance, convergence speed, and scalability for complex driving environments. . (5 Marks)
6. A game-playing AI is implemented using **DQN**, **Double DQN (DDQN)**, and **Dueling DQN**. Analyze the advantages of DDQN and Dueling DQN over the standard DQN architecture. . (5 Marks)
7. A smart drone navigation system uses **A2C** and **A3C** algorithms. Analyze the impact of parallel learning on convergence speed, computational efficiency, and policy stability. . (5 Marks)
8. A healthcare decision-support system operates with incomplete patient information. Analyze how a Partially Observable Markov Decision Process (POMDP) improves decision-making compared to a standard MDP. . (5 Marks)
9. A robotic navigation system is trained using PPO, TRPO, and DDPG. Evaluate the suitability of each algorithm for continuous control tasks based on stability, sample efficiency, and computational complexity. . (5 Marks)
10. A smart city traffic management system uses **Multi-Agent Reinforcement Learning (MARL)** to coordinate traffic signals. Evaluate the **effectiveness of MARL** in improving traffic flow while considering scalability, coordination, fairness, and ethical challenges. . (5 Marks)

Code of Conduct:

*I Dasari Haritha(192425254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



ANALYTICAL PROBLEM-SOLVING ASSESSMENT

Q3. SARSA Algorithm for Delivery Robot

Scenario

A **delivery robot operates in an uncertain environment** where obstacles or environmental conditions may change. The robot must learn a suitable path to deliver a package safely to its destination.

We can use the **SARSA (State-Action-Reward-State-Action)** algorithm because it learns from the robot's actual actions and adapts when the environment changes.

1. RL Model

Component	Delivery Robot
Agent	Delivery robot
Environment	Warehouse/road with obstacles
State	Robot's current position and surroundings
Actions	Up, Down, Left, Right
Reward	Positive for reaching destination, negative for obstacles/long paths
Algorithm	SARSA

2. State and Actions

Suppose the robot moves in a grid:

+---+---+---+---+

| S | | | |

+---+---+---+---+

| | X | | |

+---+---+---+---+

| | | | G |

+---+---+---+---+

S = Starting position

G = Goal

X = Obstacle

Possible actions:

A = {UP, DOWN, LEFT, RIGHT}

The robot observes its current state and chooses an action.

3. Reward Function

We define rewards as:

Reach destination → +10

Move to normal position → -1

Hit obstacle → -10

Invalid movement → -5

This encourages the robot to **reach the destination quickly while avoiding obstacles.**

4. SARSA Algorithm

SARSA updates the Q-value using:

Where:

- s = current state
- a = current action
- r = reward received
- s' = next state
- a' = next action
- α = learning rate
- γ = discount factor

The important point is that **SARSA uses the actual next action selected by the robot.**

5. Numerical Example

Assume:

Current state: $s = S1$

Current action: $a = \text{RIGHT}$

Reward: $r = -1$

Next state: $s' = S2$

Next action: $a' = \text{DOWN}$

Current $Q(S1, \text{RIGHT}) = 2$

$Q(S2, \text{DOWN}) = 5$

Learning rate $\alpha = 0.5$

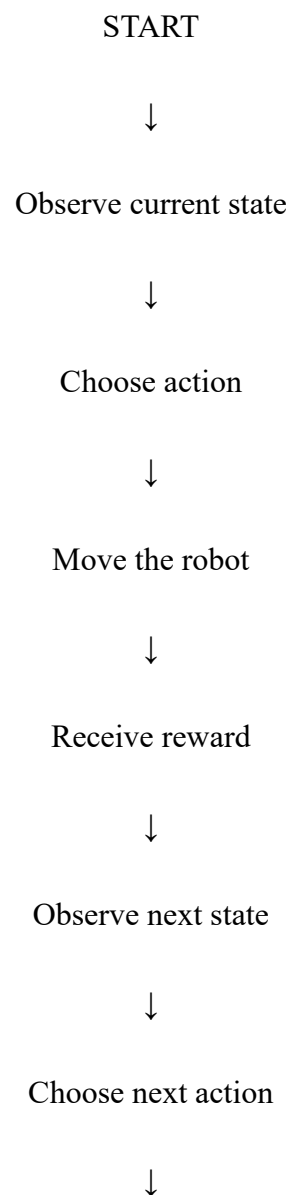
Discount factor $\gamma = 0.9$

Using SARSA:

Therefore, the value of taking **RIGHT** from state S1 increases from **2 to 2.75**.

6. How the Robot Learns

The learning process is:



Update Q-value



Repeat



Reach destination

7. How SARSA Adapts to Environmental Changes

Suppose the robot has learned this path:

Start → Right → Right → Down → Goal

Later, an obstacle appears in the path.

The robot receives a **negative reward**, for example:

Obstacle encountered → -10

The Q-value of that action decreases.

The robot then learns to select another action:

Old path:

Start → Right → Right → Down

New learned path:

Start → Right → Down → Right → Goal

Therefore, the policy adapts to the changed environment through continuous learning.

8. Why SARSA is Suitable

SARSA is suitable for this delivery robot because:

1. It learns directly from interaction with the environment.
2. It considers the **actual next action** selected by the robot.
3. It can learn safer paths in uncertain environments.
4. It can adapt when obstacles or conditions change.
5. It balances exploration and exploitation.

9. SARSA vs Q-Learning — Important Point

For this scenario, remember this difference:

SARSA	Q-Learning
On-policy	Off-policy
Uses actual next action	Uses best possible next action
Learns safer behavior in some uncertain environments	More aggressively seeks maximum estimated value
Formula uses $Q(s',a')$	Formula uses $\max Q(s',a')$

SARSA:

10. Final Answer / Conclusion

The SARSA algorithm can be used to train a delivery robot to navigate an uncertain environment. The robot observes its current state, selects an action, receives a reward, observes the next state, and selects the next action. The Q-value is then updated using the SARSA equation. When the environment changes, such as when a new obstacle appears, the robot receives negative rewards and updates its Q-values. Over repeated interactions, it learns a safer and more effective navigation policy.

 **Remember for the exam:**

SARSA = State → Action → Reward → State → Action

$S \rightarrow A \rightarrow R \rightarrow S' \rightarrow A'$

↓

Q-value Update

↓

Better Robot Policy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated

Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively